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FINAL EXAMINATION PROJECT

Phase 1

CropTrace Diagram Explanation

The business process begins when the farmer or supplier delivers crops, which in this project is called a "batch." A batch represents a specific quantity of a crop, like maize or beans, harvested at a specific time. Even if two batches are the same crop type, they are given different ID because they may have different moisture average or spoilage. This process involves four actors. The supplier who delivers the crops, the warehouse clerk who receives and records them, the quality inspector who tests the crops, and the warehouse manager who makes decisions on rejected batches.

When the batch arrives, it is received by the warehouse clerk. The clerk physically inspects the delivery and immediately uses the system to create a new record in the database, entering the batch ID, crop type, and initial yield weight. Once the physical batch is registered in the system, it is handed over to the quality inspector.

The quality inspector tests the batch to determine the spoilage and moisture data. The inspector then uses the system to create a quality check record, entering the spoilage_loss_% and moisture_avg. At this point, the database system will then take over. It automatically checks if the spoilage of the batch is greater than the maximum acceptable spoilage.

If the spoilage is greater than the threshold, the system will raise an alert. The warehouse manager receives this alert, reviews the batch, and rejects it. The system ends the process. However, if the spoilage is less than the threshold, the batch is accepted. The warehouse clerk assigns a physical storage location, and then system will update the database to record the new location and the current stock levels. The process ends successfully with a batch stored.

To conclude, this business process supports management decision making and daily operations. By including spoilage threshold in our database, it helps the system to prevent the company from storing and selling poor crops, which protects the company's reputation and reduces financial loss. Additionally, because every step is recorded in the database, management becomes clear, centralized audit. They can see exactly which supplier delivered a bad batch, how much stock is currently in the warehouse, and how quickly the clerks and inspectors are processing deliveries.